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Infectivity of PRRS virus in pig manure at different temperatures

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ABSTRACT

PRRSv is an economically important swine pathogen which can be disseminated from infected pig herds via movement of contaminated manure. The process of manure handling and inadequate cleaning of transport vehicles are commonly implicated as sources of PRRSv transmission. Stability of PRRSv in pig manure at different temperatures is unknown. The objective of this study was to determine PRRSv-infectivity half-life in manure and in a cell culture medium at 4, 20, 60 and 80 °C. To assure sample consistency across the study, all samples were prepared from common homogenized solutions (MEM and manure) and frozen at −20 °C. Samples were thawed, transferred to a water bath set at a specific temperature, inoculated with 100 μl of PRRSv at designated time points and then tested for virus infectivity. Regression models were created to estimate PRRSv half-life based on incubation temperature. There was an exponential decrease in PRRSv infectivity with increasing temperature. At every temperature tested, PRRSv had shorter half-life when incubated in manure compared to MEM. PRRSv half-life in MEM and manure was estimated at 112.6 and 120.5 h at 4 °C, 14.6 and 24.5 h at 20 °C, 1.6 and 1.7 h at 40 °C, 2.9 and 8.5 min at 60 °C, and 0.36–0.59 min at 80 °C, respectively. Results of this study can be used as basis for developing strategies to inactivate PRRSv present in manure-contaminated environments using heating treatments. For example, these data suggest that submitting transport trailers to temperature of 50 °C for 8 h would decrease PRRSv from 10^6 $TCID_{50}$/ml to less than 10^1 $TCID_{50}$/ml.

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1. Introduction

Porcine reproductive and respiratory syndrome virus (PRRSv) is an economically important pathogen in the swine industry (Neumann et al., 2005; Nieuwenhuis et al., 2012). Field reports indicate that PRRSv transmission into pig herds in North America is common (Holtkamp et al., 2010; Lambert et al., 2011; Loula, 2011). Efforts are being made to better understand PRRSv stability in the environment (Jacobs et al., 2010), as well as mechanisms of PRRSv area spread (Lambert et al., 2011; Torremorell et al., 2004). PRRSv has a long half-life time ($T_{1/2}$) in cold temperatures (Benfield et al., 1992; Bloemraad et al., 1994; Jacobs et al., 2010) such that the virus can remain infectious for extended time during winter months, facilitating indirect viral transmission (Bigras-Poulin et al., 2007; Dee et al., 2004c).

A mechanism by which PRRSv can be potentially disseminated in a region is via contaminated pig manure, either during the process of collecting manure and incorporating it in soil as fertilizer or when using transport trailers contaminated with PRRSv. The minimum infectious dose of PRRSv in manure is unknown but for aerosol exposure, minimum infectious dose for MN184, a highly transmissible strain, was estimated to be about $10^{0.25}$ $TCID_{50}$/ml (Cutler et al., 2011). In addition, Dee et al. (2005a,b,c) reported that PRRSv MN-30100 isolate remained infectious for up to 8 days in effluent from a pig farm lagoon at an average temperature of 4 °C (Dee et al., 2005c). For the purpose of this study, we define

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"pig manure" as pig faeces that can be found on the pen floor of commercial pig farms. This material is often mixed with other materials such as urine, oral fluids, feed, water and environmental dust.

While effective methods exist for cleaning contaminated transport vehicles (Dee et al., 2004a,b,c, 2005a,b), recent field data indicated that 25–50% of vehicles used for transporting pigs were not washed between loads (Bigras-Poulin et al., 2007; Lambert et al., 2012; Schneider et al., 2011). This is probably in part because of the high cost and complex logistics associated with trailer wash and disinfection. Therefore, there is a need for alternative, inexpensive methods to disinfect trailers.

Two important knowledge gaps must be elucidated to understand PRRSv indirect transmission from contaminated pig manure and methods to inactivate it: (a) PRRSv stability at moderately elevated temperatures and (b) PRRSv stability in manure samples compared to that of cell culture media (which is the best case scenario) at different ambient temperature conditions. This study evaluated half-life of PRRSv (i.e. time necessary to decrease virus infectivity by half) in manure samples compared to that of PRRSv in minimum essential medium (MEM) at 4, 20, 40, 60 and 80 °C. Information from this study is pivotal to characterize PRRSv ecology in manure samples and determine the risk of PRRSv transmission via manure. Results from this study will facilitate development of science-based strategies to inactivate PRRSv present in manure and reduce the risk of PRRSv indirect transmission.

2. Materials and methods

2.1. Virus and cell lines

A recently described field isolate of PRRSv was used for this study. The virus was characterized as restriction fragment length polymorphism (RFLP) pattern 1-18-2 belonging to Leung's Lineage 1, sublineage 5 group (Shi et al., 2010). This isolate was shown to be highly transmissible in experimental settings under field conditions (Linhares et al., 2012).

Marc-145 cell line (Kim et al., 1993) was used to propagate the virus. The third passaged virus was used in the experiment. Briefly for virus propagation, PRRSv isolate were incubated on confluent Marc-145 cells grown with cell growth medium (MEM[1] supplemented with 8% FCS[2] and antibiotics[3]) in 75 cm^2 cell culture flasks[4] at 37 °C in a humidified 5% CO_2 incubator. Following 2 h of virus-cell incubation, 50 ml of cell growth medium was added to the flask, which was returned to the incubator. On the third day of incubation, PRRSv-CPE was evident in the Marc-145 monolayer. The flask was frozen and thawed 3 times and was centrifuged at 3000 rpm for 10 min to pellet the cell debri. The supernatant containing PRRSv at 10^6 $TCID_{50}$/ml was collected in a 50 ml tube, briefly vortexed, aliquoted to 100 μl portions and stored at −80 °C freezer until use.

2.2. Pig manure

A fresh 2 kg sample of manure was obtained from two sources labelled as "a" and "b". Source "a" was a pen floor of eleven serologically PRRSv free 8-week-old pigs housed at the University of Minnesota's animal research units (Saint Paul, MN, USA). Source "b" was from a pen floor of 20–22 week-old gilts from a commercial farm. Manure samples were homogenized, kept moist and divided in aliquots of 0.90 g and placed in sterile 10 ml glass bottles with polypropylene screw caps. The pH of the homogenized manure sample was 7.5. To prevent decomposition over time, the manure aliquots were kept frozen at −20 °C until used.

2.3. Experimental design

Half-life of PRRSv infectivity in manure samples was compared to that of samples in cell culture medium (MEM) at different ambient temperatures. All samples were stored at −20 °C in 10 ml screw-cap glass bottles with 0.9 g manure or 0.9 ml MEM. To assure sample consistency across the study period, all samples were prepared from common homogenized solutions (MEM and manure), frozen at −20 °C and thawed prior to each experiment. Samples were thawed and transferred in duplicates to a water bath set at specific temperatures and inoculated with 100 μl of aliquots of PRRSv (thawed as needed from −80 °C) at designated time points (Table 1) and submitted to virus titration (expressed by $TCID_{50}$/ml) as described below.

2.4. PRRSv isolation pre-processing steps

All samples followed the same virus isolation/titration procedure. The first step of PRRSv titration was to dilute the sample by adding 3 ml of cell growth media, which facilitated the separation of virus from organic materials. Samples were then vortexed (speed 9, 20 s). Then, 2 ml of the sample were transferred to a sterile 2 ml microcentrifuge tube and centrifuged for 5 min at 10,000 rpm (9279.4 g) to separate the pellet organic debris from the supernatant. The supernatant was collected and filtered

Table 1
Sampling schedule at each temperature.

Sample #	Temperature (°C)				
	4	22	43	63	82
1	0 h	0 h	0 min	0 min	0 min
2	24 h	30 h	90 min	3 min	2 min
3	168 h	72 h	150 min	12 min	8 min
4	432 h	82 h	200 min	18 min	10 min

[1] MEM Eagle's 1X w/Earle's salts and L-glutamine, Mediatech, Inc., Manassas, VA, Cat # 10–010-CM.
[2] Fetal Calf Serum, Thermo Scientific HyClone, Logan, UT Cat # SH3085502.
[3] Mediatech, Inc., Manassas, VA, Cat # 30–001-CI.
[4] BD Falcon 75 cm^2 cell culture flask, 250 ml, tissue-culture treated polystyrene, canted neck, vented cap, Becton, Dickinson and Company, Franklin Lakes, NJ, Cat # 353136.

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using a 0.2 μm syringe filter[5] to minimize bacterial contamination. Thereafter, 100 μl of each sample was submitted to microtitration as described below.

2.5. PRRSv titration from MEM samples

Confluent Marc-145 cells monolayers in 96-well plates[6] were used for microtitration infectivity assays. Each sample was serially diluted 10-fold in cell growth media (8% FCS MEM) and 6 wells were inoculated with 100 μl of each dilution. Plates were then incubated at 37 °C in a humidified 5% CO_2 incubator and PRRSv-CPE were observed daily for 10 days. Each plate contained 6 wells of non-PRRSv-infected samples to serve as negative control. Each set of plates had the same stock PRRSv (100 μl aliquots) titrated to serve as positive control.

2.6. Statistical methods

PRRSv infectivity titers were calculated based on the probit-function method (Kim et al., 1993) using proc probit of SAS 9.2 software (SAS Institute Inc., Cary, NC, USA), and expressed as $TCID_{50}$/ml.

PRRSv infectivity half-lives in MEM and manure were calculated using a regression model with time in the log scale as described elsewhere (Govindarajulu, 2001) using proc reg of SAS 9.2 software.

3. Results

No CPE was evident in the negative control cell monolayers 10 days after sample-inoculation. An exponential decrease of PRRSv infectivity with increasing ambient temperature in both MEM and manure environment was observed (Fig. 1). Stability of PRRSv in MEM was consistently slightly longer than in manure at all temperatures. PRRSv half-life infectivity in MEM and manure respectively were 112.6 and 120.5 h at 4 °C; 14.6 and 24.5 h at 20 °C; 1.6 and 1.7 h at 40 °C; 2.9 and 8.5 min at 60 °C; and 0.36 and 0.59 min at 80 °C (Table 2).

The two linear regression models created to predict PRRSv infectivity $T_{1/2}$ from 4 to 80 °C at MEM and manure environments were: $T_{1/2} = \exp^{(9.75-0.124\times temp)}$ and $T_{1/2} = \exp^{(9.617-0.132\times temp)}$ respectively (Fig. 2).

4. Discussion

Knowledge of pathogen stability outside the host is pivotal to the understanding of indirect transmission events and consequently to allow development of approaches to prevent transmission by pathogen inactivation. PRRSv survival in the environment has been reported with virus in cell culture media or in lagoon effluent (Ajariyakhajorn et al., 1997; Benfield et al., 1992; Bloemraad et al., 1994; Dee et al., 2005c; Jacobs et al., 2010). However, when PRRSv indirect transmission occurs via fomites or transport trailers, it is important to understand the relevance of PRRSv-contaminated pig manure under field conditions. Moreover, there is no report of PRRSv half-life infectivity above 56 °C. To inactivate PRRSv in manure using temperature, knowledge of viral survival at moderately elevated temperatures is necessary. This study addressed those gaps by generating, for the first time, PRRSv survival curves in manure compared to that of in cell culture media at different ambient temperatures. In addition, we focused on measuring infectivity using virus isolation rather than quantitative PCR-based assays since RNA seems relatively stable at rising temperatures and PCR positive samples may not contain infectious virus (Zimmerman et al., 2010). The exponential decrease of PRRSv infectivity as a function of time and increasing temperature has been reported previously. One study reported exponential decrease of PRRSv infectivity in cell culture media from 4° to 30° (Jacobs et al., 2010) and another study showed a similar pattern with PRRSv exposed to up to 56 °C (Bloemraad et al., 1994). Moreover, PRRSv isolation (in CL2621 cells) from experimentally inoculated slurry samples at 4 °C, 25 °C and 37 °C has been reported for up to 14 days, 1 day and 6 h after inoculation, respectively (Ajariyakhajorn et al., 1997). However, to the best of our knowledge, this is the first study that evaluates PRRSv infectivity half life at temperatures from 4 °C up to 80 °C in manure samples. Estimations of PRRSv half-life available in the literature are summarized in Table 3.

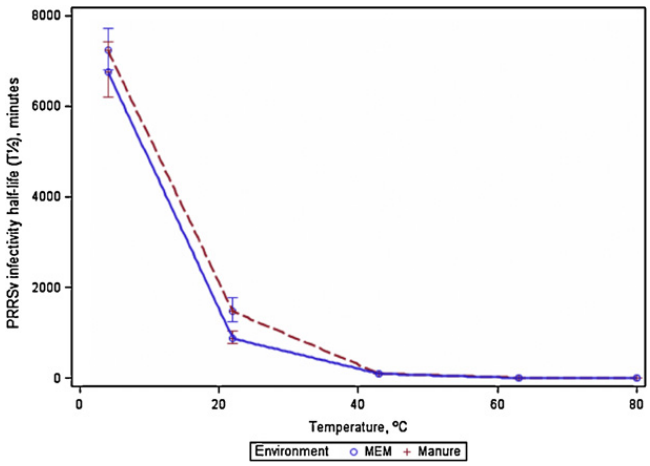

Fig. 1. Half-life PRRSv-infectivity in minutes from 4 to 80 °C in manure and cell growth media (MEM) environments.

Table 2
Median PRRSv infectious half-life and its 95% confidence interval at different ambient temperatures in cell culture media and in manure.

Temperature (°C)	$T_{1/2}$ in MEM	$T_{1/2}$ in manure
4.0	120.5 h (113.3, 128.6)	112.6 h (103.2, 123.8)
22.0	24.5 h (20.9, 29.5)	14.6 h (12.6, 17.2)
43.5	1.7 h (1.5, 1.9)	1.6 h (1.5, 1.7)
63.0	8.5 min (7.5, 9.9)	2.9 min (2.1, 4.4)
80.0	0.59 min (0.41, 1.05)	0.36 min (0.30, 0.45)

[5] Sterile Syringe Filter, Corning Inc., Corning, NY, Cat # 431215.
[6] Corning 96 well clear flat bottom polystyrene TC-treated microplates, individually wrapped, with lid, sterile, Corning Inc., Corning, NY, Cat #3596.

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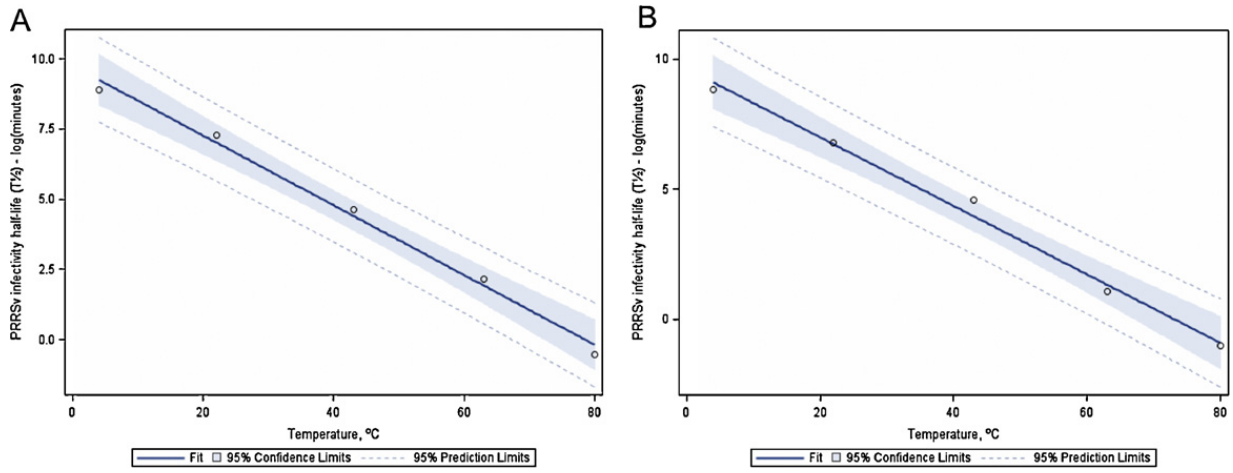

Fig. 2. PRRSv-infectivity half-life in MEM (panel A) and in manure (panel B) environments. *X*-Axes are temperature in Celsius and *Y*-axes are $\log_{10}$ of time in minutes. The solid line is the slope of the linear regression model of half-life in minutes in the log-scale. The bands around the solid line represent 95% confidence interval of the $T_{1/2}$ means (open circles). The dotted lines are the 95% prediction limits of the model. The regression model formulas for PRRSv-infectiviy half-life in MEM and manure are $T_{1/2} = \exp\ (9.747 - 0.124 \times \text{temperature})$ and $T_{1/2} = \exp\ (9.617 - 0.132 \times \text{temperature})$ respectively. These formulas predict PRRSv $T_{1/2}$ in minutes given temperature in Celsius.

Table 3
Literature reports of PRRSv infectious half-life at different ambient temperatures.

Temperature (°C)	Half-life and environment
4	140 h in cell growth media[a]
	155.5 h in cell growth media[b]
10	84.8 h in cell growth media[b]
20	27.4 h in cell growth media[b]
21	20 h in cell growth media[a]
30	1.6 h in cell growth media[b]
37	3 h in cell growth media[a]
	2.4 h in cell growth media[c]
56	6 min in cell growth media[a]
	2.3 min in cell growth media[c]

[a] Bloemraad et al. (1994).
[b] Jacobs et al. (2010).
[c] Benfield et al. (1992).

Based on the regression models generated in this study, the time necessary to inactivate PRRSv at a given temperature, or vice versa can be calculated. For instance, submitting a 10^6 $TCID_{50}$/ml pathogen to 20 half-lives would decrease its infectivity to 10^{-1} $TCID_{50}$/ml. Our model predicts it will take 3.6 months at 5 °C for PRRSv to complete 20 half-lives in manure samples (or one of the following: 15 days at 20 °C, 1.1 days at 40 °C, 7 h at 50 °C or 30 s at 70 °C).

For purposes of interpreting the $T_{1/2}$ curves beyond this study, the regression model predicted by this study assumes that the sample contaminated with PRRSv is already at a given temperature at the time when the trial starts. For practical purposes, the time it takes for an air space or a surface to reach a targeted temperature needs to be taken in consideration in addition to the treatment time. Therefore the overall treatment time should include both, the time required to reach the desirable temperature and the time it takes to inactivate the virus at that temperature. This is especially important if there are significant volumes of manure or when conditions are not conducive to reach even temperature throughout manure.

Other conditions such as humidity can also affect PRRSv infectivity (Cutler et al., 2011) and therefore the survival of PRRSv in manure. In this study, manure samples were kept saturated with humidity during the incubation time, which mimic survival estimates at the worst-case scenario. However, PRRSv infectivity in manure samples may differ if manure samples are dry or stored in conditions different from this study. These conditions need to be taken in consideration when interpreting these results and when implementing control measures under field conditions. Additionally, differences in PRRSv infectivity in manure may differ among PRRSv isolates. Differences in infectivity and transmissibility have been described for PRRSv before (Cho et al., 2006, 2007). Differences in half-life survival in pig manure environments has been reported among strains of classical swine fever virus (Weesendorp et al., 2008). However, similar half-lives values between 4 distinct PRRSv isolates were recently observed when inoculated in cell culture media at temperatures from 4 to 30 °C (Jacobs et al., 2010). Overall, strain variability did not appear to have a significant effect on infectivity when submitted to increasing ambient temperatures which suggest that strain variability may not have a significant effect on PRRSv stability in manure either. However additional studies may be needed.

To prevent PRRSv infection via transport, different approaches have been documented to successfully inactivate the virus from trailer surfaces. These protocols are all based on cleaning and power washing the trailer followed by completely drying the trailer for 8 h disinfection of the trailer with 7% glutaraldehyde and 26% quaternary ammonium chloride product (Dee et al., 2004a,c, 2005a,b); a procedure that requires mixing the disinfectant with anti-freezing solution in winter months (Dee et al., 2004a,c); and thermo-assisted drying and decontamination (TADD), which consists of using heating devices to raise the trailer temperature to about 71 °C for 30 min and allowing 120 min of rest (Dee et al., 2005a). All these procedures appear to be efficient at inactivating PRRSv

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present in trailers, but they all have important drawbacks. First, significant investments are required to build effective truck washing stations. A washing station costs between $100,000 to $150,000 USD per bay. Variable costs are about $300–$450 USD per wash.[7] Additionally, risk of cross contamination between trucks being washed in the same facility and lack of worker compliance to properly follow agro-industry biosecurity standards have been described (Racicot and Wogalter, 1995; Racicot et al., 2011) which can lead to poor trailer washing and disinfection, and subsequent dissemination of pathogens. Therefore, there is a need to test cost-effective alternatives to reliably inactivate PRRSv and other significant pathogens from transport trailers. Based on the results of this study, submitting trailers to moderately elevated temperatures of 50 °C for up to 8 h would inactivate PRRSv. This temperature and duration is sufficient to decrease virus concentration from 10^6 $TCID_{50}$/ml to virtually non-infectious via oral route (10^1 $TCID_{50}$/ml) (Dee et al., 2004c; Hermann et al., 2005).

In conclusion, this study reports for the first time, the half-live curves of PRRSv infectivity in manure at diverse ambient temperatures, ranging from cold to hot. A virus isolate previously reported to be highly virulent and transmissible between pig populations was used and prepared at relatively high concentrations. During treatment (time and temperature), samples were kept in a closed environment to conserve humidity. Experiments were performed in duplicates and results were combined and analysed in a standardized statistical package (SAS 9.2) with methods previously validated. All these conditions contributed to generate a reliable regression model representing the worst-case scenario PRRSv-survival curve. Results can be used to understand conditions of PRRSv indirect transmission as well as the development of methods to inactivate the virus using temperature treatments.

Competing interests

None.

Authors' contributions

DL, HSJ, MT, RB contributed in experiment design; HSJ contributed in lab expertise; DL contributed in execution of experiment; DL, MT, RB contributed in data analysis; DL, MT, MT, RB contributed in manuscript preparation.

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[7] Loula and Schwartz, personal communication, 2011.

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